



Ground Penetrating Radar (GPR) and EM Loupe for Geophysical Surveys in Mining

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Executive Summary: This project investigated the use of Ground Penetrating Radar (GPR) and EM Loupe transient electromagnetic surveying for near-surface geophysical investigation at the Costerfield survey site. The study aimed to evaluate the effectiveness of integrating shallow GPR reflection profiling with EM conductivity-depth imaging to improve preliminary interpretation of subsurface conditions within a mining environment. Field surveys were completed using a pulseEKKO 50 MHz GPR system and the EM Loupe transient electromagnetic system. Data processing and interpretation were carried out using EKKO Project, EmaxAIR and OriginPro software.

The GPR survey identified shallow reflection boundaries and possible fault-like disturbed zones within the upper approximately 10 m of the subsurface. Several large hyperbolic reflections were interpreted as air-wave interference rather than genuine buried targets. The EM Loupe survey provided conductivity information to depths of approximately 50 m and identified a progressive decrease in conductivity with depth across the survey area. High conductivity values within the shallow subsurface were interpreted as moist clay-rich or saturated fine-grained material, while lower conductivity values at greater depths were interpreted as less weathered Costerfield Siltstone containing minor sandstone interbeds. The integrated interpretation of GPR reflection boundaries and EM conductivity variation improved confidence in the interpretation of shallow subsurface layering and structural variability.

The project demonstrated that the combined use of GPR and EM Loupe can provide useful preliminary subsurface information across different investigation depths within an active mining environment. GPR was effective for identifying shallow structural features, while EM Loupe provided broader conductivity-depth trends and spatial conductivity variation across the survey area. However, the interpretations remain preliminary because no direct subsurface validation data, such as borehole logs or drillhole information, were available. The project concluded that the combined use of non-intrusive geophysical methods together with intrusive investigation methods has strong potential to improve future mining site investigation and geological assessment.

1 Introduction: geophysical surveys in mining

Near-surface geophysical methods are widely used in mining, engineering and environmental investigations to improve understanding of subsurface conditions without extensive excavation or drilling. Ground Penetrating Radar (GPR) is commonly used for shallow high-resolution imaging of subsurface reflectors, fractures and structural boundaries, while transient electromagnetic (TEM) methods are effective for mapping conductivity variation associated with groundwater, weathering and geological material changes. The integration of multiple geophysical methods can improve interpretation confidence by combining different types of subsurface information.

The Costerfield mining area in Victoria is characterised by structurally controlled gold-antimony mineralisation associated with the Costerfield Siltstone geological unit. Existing geological information suggests the presence of weathered overburden material above thin-bedded siltstone with minor sandstone interbeds. However, shallow subsurface conditions across the survey area remain incompletely understood. This project investigated the use of integrated GPR and EM Loupe surveying to improve preliminary interpretation of shallow subsurface conditions at the Costerfield survey site.

The project involved field acquisition, data processing and integrated interpretation using a pulseEKKO 50 MHz GPR system and the EM Loupe transient electromagnetic system. GPR reflection profiles were used to identify shallow reflection boundaries and possible structural disturbances, while EM Loupe conductivity-depth imaging and depth slices were used to interpret conductivity variation across the survey area. The results were compared with published conductivity reference values and the local geological setting to develop a preliminary interpretation of subsurface layering and material variation.

The objectives of this project were to:

- investigate the practical application of GPR and EM Loupe surveying within a mining environment;
- identify shallow subsurface boundaries, conductivity variation and possible disturbed zones;
- evaluate the effectiveness of integrating GPR and EM conductivity data for preliminary subsurface interpretation; and
- assess the limitations and uncertainties associated with near-surface geophysical investigation at the Costerfield survey site.

1.1 Near-surface geophysical investigation in mining

Near-surface geophysical investigation is important in mining because shallow ground conditions can affect site planning, excavation safety, groundwater control, foundation design, and environmental management (Butler, 2005; Reynolds, 2011). In mining and geotechnical contexts, features such as soil layering, regolith, weathered rock, faults, fractures, groundwater pathways, voids, and conductive zones may influence both operational safety and infrastructure performance (Dentith & Mudge, 2014; Sharma, 1997). These features are also relevant to tailings storage facilities, waste dumps, haul roads, and water storage areas, where seepage, salinity, weathering, and bedrock topography may affect design and monitoring decisions (Loupe Geophysics Pty Ltd., 2024; Reynolds, 2011).

Traditional investigation methods, including boreholes, test pits, sampling, and cone penetration tests, remain essential because they provide direct observations of ground conditions and geotechnical properties (Clayton et al., 1995). However, boreholes and test pits only describe conditions at individual investigation points, so lateral changes between points may be missed when the subsurface is heterogeneous (Clayton et al., 1995; Reynolds, 2011). CPT can provide a continuous vertical profile, but it still represents a narrow column of ground and therefore requires multiple tests to characterise spatial variability across a wider site (Clayton et al., 1995; Butler, 2005). Drilling and excavation can also be costly and time-consuming, especially in active mine sites where access, safety control, and ground disturbance must be carefully managed (Sharma, 1997; Reynolds, 2011). For this reason, geophysical surveys are often used as a complementary approach because they can provide continuous subsurface information between sparse investigation points (Butler, 2005; Reynolds, 2011). Together, these studies indicate that near-surface geophysical methods are most useful in mining when they are used alongside direct investigation methods. Drilling, test pits and CPTs provide reliable ground-truth information at specific points, while geophysical surveys help extend this information across the wider site by improving spatial continuity between limited investigation points.

This project focuses on Ground Penetrating Radar and the EM Loupe system because they provide different but complementary types of subsurface information. This selection is directly related to the conditions at Costerfield Site 3B, where the near-surface profile includes alluvial cover of uncertain thickness, possible shallow structural variation, and conductivity changes that may be associated with groundwater, weathering, or mineralisation-related materials. GPR is commonly used to identify shallow reflection features, buried targets, and subsurface interfaces, especially where changes in material properties create clear radar responses (Butler, 2005; Sensors & Software Inc., 2022). In contrast, EM Loupe, as a portable time-domain electromagnetic system, is used to map near-surface conductivity variations and can support the identification of conductive zones related to groundwater, salinity, weathering, tailings seepage, or conductive geological materials (Loupe Geophysics Pty Ltd., 2024). Therefore, these two methods are reviewed together to support the integrated GPR–EM Loupe survey approach used in this project.

1.2 Ground Penetrating Radar (GPR): principles and applications

1.2.1 Ground Penetrating Radar (GPR): Basic Principles

Ground Penetrating Radar (GPR) is a non-destructive geophysical method that uses high-frequency electromagnetic waves to investigate shallow subsurface structures. In a typical survey, a transmitting antenna emits electromagnetic pulses into the ground, and reflections are generated when the waves encounter contrasts in permittivity, electrical conductivity or other electromagnetic properties. The receiving antenna records the two-way travel time, amplitude and waveform of the returned signals, which are then processed to produce subsurface radar profiles. Davis and Annan (1989) describe GPR as an electromagnetic method analogous to reflection seismology, with imaging quality controlled mainly by the high-frequency electrical properties of subsurface materials.

The propagation velocity of radar waves in the subsurface medium is primarily governed by the relative permittivity and can be expressed as:

$$v = \frac{c}{\sqrt{K}}$$

Here, v is the radar-wave velocity, c is the speed of light in vacuum, and K is the relative permittivity. Because water has a much higher permittivity than most dry soils and rocks, variations in moisture content, lithology and fractures can strongly affect radar velocity and reflection responses. Therefore, GPR is sensitive to layer boundaries, water-filled fractures, shallow groundwater changes and lithological contrasts (Davis & Annan, 1989).

$$D = \frac{VT}{2}$$

Here, D is target depth, v is radar-wave velocity, and T is two-way travel time. This shows that raw GPR data are time-based and must be converted into geologically meaningful depth profiles using appropriate velocity assumptions or field calibration.

GPR reflectivity depends primarily on the electrical properties of the media on either side of the subsurface interface. In simplified terms, the reflectivity coefficient can be expressed as:

Depth interpretation in GPR is based on the time-of-flight relationship:

$$R = \frac{\sqrt{K_2} - \sqrt{K_1}}{\sqrt{K_2} + \sqrt{K_1}}$$

Here, K_1 and K_2 represent the relative permittivities of the media on either side of the interface, respectively. This relationship indicates that the more pronounced the difference in the dielectric properties of the two media, the clearer the reflection in the GPR profile. Consequently, fractures, voids, lithological boundaries, changes in overburden, and water-bearing anomalies are generally more easily identified by GPR.

The choice of frequency for GPR is always a trade-off between penetration depth and resolution. The relationship between radar wavelength and frequency is given by:

$$\lambda = \frac{v}{f}$$

Where λ is the wavelength, v is the speed of the radar wave, and f is the antenna frequency. The vertical resolution of GPR can usually be approximated as:

$$\Delta z \approx \frac{\lambda}{4}$$

Consequently, low-frequency antennas typically offer greater penetration capability but lower resolution; high-frequency antennas provide more detailed information at shallow depths but have a limited effective detection depth. Francke (2012) points out that, in mineral resource evaluation, the key to GPR lies not in simply pursuing the greatest detection depth or the highest resolution, but in striking a reasonable balance based on the target strata, the nature of the overburden and site conditions.

Furthermore, GPR signals gradually attenuate with increasing propagation depth; this attenuation is particularly pronounced in strata with high clay content, high water content or high electrical conductivity, thereby reducing the effective detection depth and imaging quality. This attenuation process can be conceptually represented as:

$$A(z) = A_0 e^{-\alpha z}$$

Hence, $A(z)$ represents the signal amplitude at depth z , A_0 is the initial amplitude, and α is the attenuation coefficient. This relationship illustrates that even if a reflective interface exists underground, deep structures may not be clearly visible in the radar profile if signal attenuation is too severe. This trade-off is directly related to Costerfield's survey work. As alluvial cover was anticipated at Site 3B and the shallow structural conditions were unclear, a 50 MHz antenna was selected to achieve greater penetration depth, even though this would reduce the ability to resolve fine-scale features near the surface.

1.2.2 The advantages of GPR in Mining

The main advantages of GPR in mining applications lie in its high resolution, continuous imaging, non-destructive nature and high on-site efficiency. Unlike drilling, which can only provide information at discrete points, GPR can generate continuous subsurface profiles along a survey line, thereby aiding the assessment of stratigraphic continuity between boreholes, changes in overburden, bedrock interfaces, fractures and shallow anomalies. This is particularly important in mines and quarries, as fractures, weathered zones, voids and structural discontinuities typically exhibit significant spatial heterogeneity, making it difficult to fully assess their distribution relying solely on a limited number of boreholes (Davis & Annan, 1989). Davis and Annan (1989) and Francke (2012) share the view that the primary value of GPR lies in filling information gaps between boreholes and helping to establish the continuity of shallow strata and structures, rather than serving as a substitute for drilling itself.

Furthermore, GPR can rapidly acquire shallow subsurface information without damaging the ground surface or surrounding rock and is therefore often used as a complementary tool for mineral resource evaluation, quarry structure identification and subsurface risk assessment. Francke (2012) noted that GPR can be used to establish stratigraphic continuity between boreholes, reducing the uncertainty associated with relying solely on borehole interpretation. In underground mines and quarries, GPR can also assist in identifying karst features, voids and other potential geological hazards (Caselle et al., 2020). However, these findings share a common premise: GPR imaging performance depends on low electrical conductivity and favourable ground conditions; if the site contains clay, water-saturated overburden or highly conductive materials, both penetration depth and image quality will be significantly reduced.

Therefore, in this project, GPR should not be regarded as a tool for directly identifying ore bodies, but rather as a rapid, continuous and non-invasive method for investigating shallow subsurface structures. Relevant studies indicate that GPR is sensitive to fractures, voids, lithological boundaries and hydrogeological anomalies, and can support risk assessment and decision-making regarding excavation safety (Baggett et al., 2020; Caselle et al., 2020). Consequently, when GPR is used in conjunction with drilling data, geological information and other geophysical methods, it can enhance the reliability of near-surface interpretation.

1.2.3 The Limitations of GPR in Mining

Although GPR offers advantages such as high resolution, continuous imaging and non-destructive operation, its effectiveness is highly dependent on the site's stratigraphic conditions. GPR is highly sensitive to the electrical conductivity of subsurface media and typically performs well in low-conductivity environments such as dry sand, gravel or relatively intact hard rock; however, in clay, silt, saline-alkali strata or environments containing conductive pore water, electromagnetic waves attenuate rapidly, thereby reducing the effective detection depth and image quality. Consequently, the

applicability of GPR in mining areas is often limited by the nature of the overburden, water-bearing conditions and the electrical characteristics of the strata (Davis & Annan, 1989).

Secondly, the interpretation of GPR data involves a degree of uncertainty. Hyperbolas, reflection interfaces, scattering bands and amplitude variations in radar profiles do not always correspond to a single geological interpretation. In quarries or complex rock mass environments, structures such as fractures, voids, bedding, schistosity and anisotropy may produce similar or overlapping radar responses; consequently, interpretation results must be evaluated in conjunction with the in-situ geological context, target morphology and other corroborating data (Rey et al., 2015).

Finally, GPR is better suited as a high-resolution imaging tool for shallow to near-surface applications rather than as a deep exploration method. Although low-frequency antennas can improve penetration, they reduce resolution and are subject to greater restrictions in field deployment. Consequently, in practical mine surveys, GPR should not be relied upon alone for comprehensive subsurface interpretation but should be used in conjunction with methods such as drilling, geological data, logging, or EM Loupe to enhance the reliability of interpretations (Francke, 2012; Lombardi et al., 2022).

1.2.4 Key applications of GPR in mining and quarrying

In mining and quarrying applications, the primary value of GPR lies not in the direct detection of deep ore bodies, but in providing continuous information on shallow subsurface structures to assist with resource assessment, overburden interpretation and the determination of stratigraphic continuity between boreholes. Francke (2012) notes that GPR has been applied in a variety of mineral resource assessment scenarios, including laterites, bauxite, iron ore, mineral sands, coal, kimberlites and palaeochannels, where its core role is typically to identify shallow overburden, bedrock interfaces, stratigraphic continuity and the shallow geometric boundaries of ore bodies. Consequently, in surface mineral resource evaluation, GPR is better suited as a complementary tool to drilling, used to reduce the spatial uncertainty associated with relying solely on borehole interpretation, rather than as a standalone method for confirming mineralisation (Francke, 2012).

In quarry and shallow rock body surveys, GPR is frequently used to identify fractures, structural discontinuities, rock mass boundaries and variations in quarry quality. In a study of decorative granite quarries, Porsani et al. (2006) demonstrated that GPR can identify fractures, unloading joints, large blocks of intact rock and fresh granite outcrops, and can provide a basis for determining quarry advance directions, reducing blasting and optimising extraction costs. Rey et al. (2015) also noted that GPR can be used to distinguish between different stone units and to identify discontinuities such as fractures, voids and anisotropy, thereby supporting reserve assessment, face design and rock mass quality evaluation. In recent years, GPR has also been further utilised for mining optimisation and sustainable resource utilisation, for example, in combination with optimisation algorithms to assess cutting directions, block sizes and recoverable volumes (Reed & Bonduà, 2025). These studies demonstrate that the role of GPR in quarries has gradually expanded from simple geological identification to mining decision-making and production optimisation. Compared with applications such as the identification of coal-rock interfaces in coal mines, these studies of quarries are more relevant to Costerfield, as they similarly focus on fractures, discontinuities, contact zones and shallow structural variations in hard rock environments.

GPR is equally valuable for underground mine safety and the identification of karst-related risks. Baggett et al. (2020) note that in underground limestone mines, karst cavities can lead to water inrushes, structural instability and engineering uncertainties, whilst GPR can identify strong reflection anomalies

by exploiting the distinct dielectric differences between limestone and open cavities. Caselle et al. (2020) further demonstrate that in underground gypsum mines, GPR is not only capable of identifying karst structures but can also assist in distinguishing between gas-filled, water-filled and clay-filled karst voids, thereby supporting risk assessment and safe excavation decision-making. Consequently, in such scenarios, the significance of GPR extends beyond mere imaging and is directly linked to early warning systems and risk management in underground mining (Baggett et al., 2020; Caselle et al., 2020).

Although GPR has also been applied in coal mining and mining-related environmental monitoring, including coal–rock interface detection, abandoned workings identification and soil moisture estimation in disturbed mining areas, these applications are less directly relevant to this project (Ma et al., 2023; Qiu et al., 2025; Turner et al., 1989; Wu et al., 2020; Xu et al., 2018). For the Costerfield gold–antimony mine survey, the focus is instead on the use of GPR for shallow structural characterisation in a hard-rock mining environment. In a structurally controlled vein-type deposit such as Costerfield, GPR should not be regarded as a direct ore-detection method, but as a tool for identifying shallow interfaces, overburden changes, fractures, contacts, structural discontinuities and possible vein-related expressions. Previous Deep Ground Penetrating Radar (DGPR) results near Costerfield also suggest that radar profiles can identify fault-like, shear-like, contact and vein-like features, although vegetation, tree roots and surface conditions may introduce noise. Therefore, in this project, GPR is most appropriately used as a line-based structural characterisation method that complements EM Loupe conductivity mapping, rather than as standalone evidence for mineralisation.

1.3 EM: principles and applications

1.3.1 EM: Basic Principles

The EM Loupe system is based on the transient electromagnetic method (TEM), also known as the time-domain electromagnetic method (TDEM) (Loupe Geophysics, 2023; Van Dam et al., 2020). Therefore, the basic TEM principle needs to be introduced before discussing the Loupe system itself, because TEM explains how the system measures subsurface conductivity, while the Loupe system represents a portable and field-efficient implementation of this method (Loupe Geophysics, n.d.; Loupe Geophysics, 2024; Van Dam et al., 2020).

In a TEM survey, a transmitter loop produces a primary magnetic field by passing an electrical current through the loop (Loupe Geophysics, 2023; Van Dam et al., 2020). When the current is switched off, the changing magnetic field induces eddy currents in conductive subsurface materials, and these currents generate a secondary electromagnetic field (Sánchez Mora et al., 2024; Van Dam et al., 2020). The receiver records the decay of this secondary field, and the recorded response is related to the electrical conductivity distribution of the ground (Loupe Geophysics, n.d.; Van Dam et al., 2020).

The time-domain response provides the basis for depth interpretation (Sánchez Mora et al., 2024; Van Dam et al., 2020). Earlier time channels mainly represent shallower responses, while later time channels contain information from greater effective depths (Street & Mayes, 2025; Van Dam et al., 2020). These responses can be transformed into apparent conductivity-depth pairs, conductivity-depth imaging sections, or inverted conductivity models, depending on the processing method and data quality (Sánchez Mora et al., 2024; Van Dam et al., 2020). The effective depth is not fixed, because it depends on transmitter moment, ground resistivity, system noise, stacking, and processing choices (Loupe Geophysics, n.d.; Sánchez Mora et al., 2024; Van Dam et al., 2020).

The EM Loupe system applies this TEM principle in a backpack-mounted moving-loop configuration (Loupe Geophysics, 2023; Soares & Wijns, 2023; Van Dam et al., 2020). In standard walking mode, one operator carries the transmitter loop and another carries the three-component receiver, while a tablet or smart device supports navigation, acquisition control, and real-time quality checking (Loupe Geophysics, 2023; Street & Duncan, 2021; Van Dam et al., 2020). The system records full time-series data and uses processing functions such as stacking, filtering, windowing, and motion-noise reduction to improve data quality in field conditions (Loupe Geophysics, n.d.; Street & Duncan, 2023; Van Dam et al., 2020). Therefore, EM Loupe should be understood as a portable TEM system for near-surface conductivity measurement rather than a separate geophysical method (Loupe Geophysics, 2024; Street & Duncan, 2021; Street & Mayes, 2025).

1.3.2 The advantages of EM in Mining

The EM Loupe system provides a non-invasive method for mapping near-surface conductivity variations without drilling or excavation (Loupe Geophysics, n.d.; Van Dam et al., 2020). This is useful in mine-site and environmental investigations because conductivity can be affected by soil texture, moisture content, clay content, groundwater salinity, and conductive geological materials (Soares & Wijns, 2023; Van Dam et al., 2020). The method can therefore provide spatial information that is difficult to obtain from sparse point-based observations alone (Sánchez Mora et al., 2024; Soares & Wijns, 2023).

A major advantage of the Loupe system is its portability and field efficiency. Unlike conventional ground-based moving-loop TEM surveys, which are often slower to deploy, Loupe is carried by two operators using backpack-mounted transmitter and receiver units (Loupe Geophysics, 2023; Van Dam et al., 2020). This design allows continuous walking-mode acquisition and makes the system suitable for uneven ground, dense vegetation, narrow access areas, and difficult terrain where vehicle-towed systems may be impractical (Street & Mayes, 2025; Van Dam et al., 2020).

The system also supports spatially continuous conductivity mapping. Loupe records full time-series data and can produce gridded conductivity maps and conductivity-depth sections after processing (Loupe Geophysics, n.d.; Sánchez Mora et al., 2024). Van Dam et al. (2020) showed that dense Loupe profiles can identify laterally coherent conductivity zones and can be transformed into conductivity-depth images for near-surface interpretation. The system also includes real-time processing and quality control functions, including stacking, filtering, windowing, and motion-noise reduction, which help improve data quality under field conditions with electromagnetic or operator-motion noise (Loupe Geophysics, n.d.; Van Dam et al., 2020).

1.3.3 The Limitations of EM in Mining

Despite these advantages, EM Loupe data require careful interpretation because conductivity anomalies are not unique. A high-conductivity response may be caused by clay-rich material, groundwater, saline pore fluids, conductive mineralisation, metal objects, contamination, or increased weathering (Soares & Wijns, 2023; Van Dam et al., 2020). For example, Soares and Wijns (2023) found that clays strongly influenced conductivity in nickel laterite profiles, but the main regolith units had overlapping conductivity ranges. This means that Loupe results cannot always identify geological boundaries without supporting geological or geotechnical information (Sánchez Mora et al., 2024; Soares & Wijns, 2023).

The effective depth and resolution of Loupe surveys are also variable. The depth of investigation depends on transmitter moment, ground resistivity, random noise, stacking, and processing quality

(Loupe Geophysics, n.d.; Van Dam et al., 2020). Loupe can provide useful near-surface conductivity-depth information, but it may not resolve small or thin features as clearly as methods with higher spatial or vertical resolution (Sánchez Mora et al., 2024; Soares & Wijns, 2023). Sánchez Mora et al. (2024) also showed that conductivity-depth imaging can provide a useful estimate of bulk conductivity, but it may lack depth accuracy compared with 1D inversion constrained by borehole data.

Field conditions can further affect data quality. Metal infrastructure, power lines, fences, vehicles, surface debris, steep topography, vegetation, and operator movement can introduce cultural or motion-related noise (Loupe Geophysics, n.d.; Van Dam et al., 2020). Although the system includes filtering and stacking functions, additional processing may still be required to remove spurious readings and improve signal-to-noise ratio (Van Dam et al., 2020). Therefore, Loupe data should be interpreted with site geology, field observations, borehole logs, CPT results, or other geophysical datasets to avoid over-interpreting conductivity anomalies (Sánchez Mora et al., 2024; Van Dam et al., 2020).

1.3.4 Key applications of EM in mining and quarrying

EM methods are widely used in mining and quarry surveys, as variations in subsurface electrical conductivity can reflect geological conditions, water content, clay content, groundwater distribution, and anomalies potentially associated with conductive mineralisation (Loupe Geophysics, 2023; Van Dam et al., 2020). In mineral exploration and geological mapping, EM surveys can be used to identify conductive zones potentially associated with mineralisation, geological structures or weathering boundaries (Street & Mayes, 2025; Van Dam et al., 2020). For example, the Loupe TEM has been used to delineate exploration targets in complex or difficult-to-access terrain, demonstrating the method's practical value in mining environments where rapid coverage and portable deployment are required (Street & Mayes, 2025). At the same time, EM methods can also be used to map weathered zones, overburden and shallow geological conditions, as variations in electrical conductivity are often related to clay content, water content and degree of weathering (Soares & Wijns, 2023). Consequently, in mining and quarrying environments, EM methods not only support the identification of potential electrical anomalies but also aid in understanding shallow overburden and geotechnical conditions, information that is closely linked to excavation planning, material assessment and site development (Sánchez Mora et al., 2024; Soares & Wijns, 2023).

Beyond exploration and geological mapping, the same conductivity information can also be utilised for groundwater investigations and environmental monitoring around mining infrastructure. Groundwater, salinity, seepage pathways and aquifer zones typically alter the conductivity of the subsurface medium; consequently, EM data can be used to assist in identifying potential seepage pathways and hydrogeological anomalies around tailings facilities, waste rock dumps, water storage areas or other mining infrastructure (Loupe Geophysics, 2023; Van Dam et al., 2020). This demonstrates that the role of EM methods in mining is not limited to resource exploration but can also serve environmental risk assessment and long-term monitoring of mining areas (Van Dam et al., 2021). However, EM conductivity anomalies are not unambiguous: higher conductivity may result from groundwater, clay, weathering, salinity, metallic interference or mineralisation. Consequently, EM results should not be regarded in isolation as direct evidence of mineralisation or specific geological structures, but should be interpreted holistically in conjunction with the geological context, field observations and other geophysical data such as GPR (Sánchez Mora et al., 2024; Soares & Wijns, 2023).

1.4 Rationale for integrated GPR–EM Loupe survey

Near-surface investigation in mining and quarrying requires data that can support reliable subsurface interpretation without disturbing the ground. The survey method should provide evidence of both shallow structure and material-property variation, because one type of data alone may not explain complex ground conditions (Sánchez Mora et al., 2024; Van Dam et al., 2020).

GPR and EM Loupe were selected because they provide different but related information. GPR can image shallow reflection features and structural boundaries, but its signal may weaken in conductive or moisture-rich ground (Davis & Annan, 1989; Solla et al., 2021). EM Loupe can map conductivity variation efficiently, but conductivity anomalies are not unique and may require geological support for interpretation (Soares & Wijns, 2023; Van Dam et al., 2020).

The main reason for integrating the two methods is their complementarity. GPR mainly responds to dielectric contrasts, so it is useful for identifying possible interfaces, layering, and discrete shallow features (Davis & Annan, 1989; Solla et al., 2021). EM Loupe mainly responds to conductivity contrasts, so it can indicate changes related to moisture, clay content, weathering, or conductive materials (Soares & Wijns, 2023; Van Dam et al., 2020). When a GPR reflector corresponds with an EM conductivity change, the interpretation becomes more reliable because two independent physical responses support the same feature. When the two datasets do not match, the difference can still help identify uncertainty, because the feature may only affect one physical property or one method may be limited by local ground conditions (Sánchez Mora et al., 2024).

Therefore, the integrated GPR–EM Loupe approach provides a stronger basis for later field interpretation than either method alone. Existing studies have demonstrated the individual value of GPR and electromagnetic methods in mining and near-surface investigations, but few have examined the combined use of GPR reflection profiling and walking-mode EM Loupe conductivity mapping at an active gold–antimony mine with complex near-surface geology.

This project addresses that gap by applying both methods through a Royal Park training survey and a Costerfield mine-site survey, testing whether integrated reflection and conductivity information can improve near-surface characterisation for practical mine-site interpretation and client decision-making

2 Methodology

2.1 Overall methodological design

This project adopted a two-phase methodological design. The first phase was conducted at Royal Park as a practice survey, and the second phase was conducted at Costerfield Mine as the final field investigation. This structure allowed the team to first test equipment operation and field procedures in a controlled setting before applying the workflow in a mine-site environment.

The Royal Park survey focused on equipment training and workflow validation. The team practised GPR and EM Loupe operation, survey-line acquisition, field note recording, real-time data checking, and data backup. This stage was not intended to provide the main geological interpretation for the project.

The Costerfield survey formed the main field application. The workflow developed through the Royal Park practice was applied to collect near-surface geophysical data under real mining conditions. This

stage provided the primary dataset for interpreting shallow subsurface conditions within the selected survey area.

GPR and EM Loupe were used as complementary methods. GPR provided line-based shallow reflection profiles for identifying subsurface interfaces and structural changes. EM Loupe provided conductivity mapping and conductivity-depth sections for assessing spatial variations in near-surface electrical conductivity. The integrated use of both datasets supported a more reliable interpretation of near-surface geological conditions.

2.2 Phase 1: Royal Park practice survey

2.2.1 Survey objective and training target

The Royal Park survey served as a preparatory field exercise before the final mine-site survey. Its main value was field training rather than geological interpretation. Through this practice, the team became familiar with the complete field workflow, including equipment preparation, site inspection, grid setup, survey-line operation, field notes, data checking, site photography, and data backup.

The practice also improved the team's field organisation. The team clarified the order of work from equipment checking at the university to equipment transport, site setup, data acquisition, file checking, and backup after each survey stage. This process helped the team manage time, roles, equipment movement, and field records more systematically during fieldwork.

For the GPR survey, the team mainly practised system operation and data quality control. The practice improved familiarity with the computer interface, file creation, data saving, display monitoring, and field note recording. When two GPR lines showed incomplete radar profiles, the issue was identified on site and the affected lines were remeasured immediately.

For the EM Loupe survey, the team mainly practised the moving-loop walking method. The key training outcomes included two-operator coordination, transmitter–receiver spacing control, line following, steady walking speed, tablet navigation, signal checking, and data saving after each line.

2.2.2 Field layout and practice grid design

The Royal Park practice survey was conducted in Parkville, Victoria. The site was a flat, grass-covered area with compacted soil and minor vegetation. These surface conditions were suitable for equipment training because they allowed safe movement of the GPR SmartCart and EM Loupe operators along repeated survey lines.

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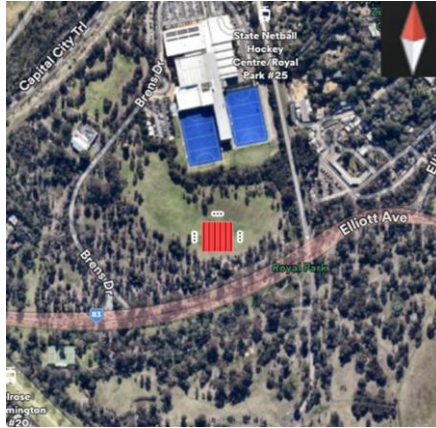


Figure 1. Royal Park practice survey location and grid layout

A $50\text{ m} \times 50\text{ m}$ square grid was established for the survey. The four corners were labelled A, B, C, and D, with Point A used as the reference point. Survey line 1 was collected from A to B, and the odd-numbered lines, including Lines 1, 3, 5, 7, and 9, followed the same A-to-B direction. The grid was set out using GPS coordinates, measuring tapes, pegs, marker ropes, and a compass. The corner angles were checked using the Pythagorean method to improve the accuracy of the grid layout before data collection.



Figure 2. Royal Park practice survey grid layout

The survey lines were arranged in the north–south direction. A 5 m line spacing was used across the grid. The lines were collected in an alternating direction, with each adjacent line surveyed in the opposite direction. This approach reduced unnecessary walking time and improved field efficiency.

The same grid was used for both GPR and EM Loupe data collection. This allowed the two datasets to share the same spatial reference and made later comparison between radar reflection data and electromagnetic conductivity data more consistent.

2.2.3 Equipment setup and practice survey method

2.2.3.1 GPR SmartCart setup and acquisition practice

In the Royal Park practice survey, GPR data were collected using a 100 MHz antenna mounted on the SmartCart system. This stage was used to practise the GPR operating workflow under simple field conditions, rather than to conduct detailed geological interpretation. The SmartCart platform was suitable for this practice because it supports line-based GPR data acquisition on relatively flat and open surfaces (Sensors & Software Inc., 2022).

Before data acquisition, the team assembled and checked the main GPR components, including the control unit, antenna, cart frame, cables, batteries, and data-recording interface. The system display was checked before the survey lines were started. Survey files were also created in advance to confirm that the GPR computer system could record and save data correctly.

During acquisition, the SmartCart was pushed steadily along each marked survey line. The team practised line alignment, stable cart movement, display monitoring, file naming, data saving, and brief field note recording after each line. These procedures helped the team become familiar with GPR system operation and field data management, although the later Costerfield survey used a different GPR configuration and survey method.

2.2.3.2 EM Loupe setup and walking-line practice

In the Royal Park practice survey, EM Loupe data were collected using a transmitter frequency of 75 Hz and a transmitter–receiver separation of 5 m. Although the later Costerfield survey adopted different settings, with a 375 Hz transmitter frequency and 10 m separation, the basic moving-loop acquisition method remained consistent. The Royal Park survey was therefore used to practise the EM Loupe field workflow, including system setup, operator coordination, line navigation, and field data management.

Before data acquisition, the team prepared the transmitter backpack, receiver backpack, connecting cable, batteries, and tablet interface. The EM Loupe moving-loop method requires two operators, with one operator carrying the transmitter system and the other carrying the receiver at a fixed separation (Loupe Geophysics Pty Ltd, 2023). The tablet interface was used to monitor navigation, system status, and real-time data display before each survey line.

During acquisition, the two operators walked along the marked survey lines while maintaining a stable walking speed and fixed transmitter–receiver spacing. The team practised line following, spacing control, real-time signal checking, file naming, data saving, and brief field note recording. These procedures were important because consistent movement and stable transmitter–receiver separation help reduce unwanted motion-related noise in EM Loupe data (Loupe Geophysics Pty Ltd, 2023).

2.2.3.3 Field workflow and data checks

The Royal Park practice survey followed a full field workflow from equipment preparation to data backup. Before travelling to the site, the team checked the GPR and EM Loupe equipment at the university and loaded the instruments into the vehicle. After arriving at Royal Park, the team inspected the survey area, confirmed the field layout, and prepared the marked survey lines before data collection.

The GPR survey was conducted first, followed by the EM Loupe survey. During the fieldwork, the team recorded the survey layout, testing lines, acquisition order, and computer parameter settings. Site

photographs were also taken to document the grid layout, equipment setup, field operation, and system settings.

After each survey stage, the team checked whether the data files had been saved successfully and whether the recorded data appeared normal on the system display. The data were then backed up to the team's own computer to reduce the risk of data loss.

The Royal Park survey also provided an opportunity to practise real-time data quality control before the final mine-site survey. During the GPR survey, two lines showed incomplete radar profiles, with approximately half of the displayed data missing. The issue was identified on site through the system display, and the two affected lines were remeasured immediately. This process helped the team practise data checking, problem identification, and repeat acquisition under field conditions.

2.2.4 Lessons transferred to the Costerfield survey

The Royal Park survey served as a preparatory field exercise before the final mine-site survey. Its main value was field training rather than geological interpretation. Through this practice, the team became familiar with the full field workflow, including equipment preparation, site inspection, grid setup, survey-line operation, field notes, data checking, photography, and data backup.

The practice also helped the team improve its field organisation. The team clarified the order of work from equipment checking at the university to site setup, data acquisition, file checking, and backup after each survey stage. This process helped the team understand how to manage time, roles, equipment movement, and field records during a complete geophysical survey.

For the GPR survey, the team mainly practised system operation and data quality control. The team became more familiar with the computer interface, file creation, data saving, display monitoring, and field note recording. When two GPR lines showed incomplete radar profiles, the issue was identified on site and the affected lines were remeasured immediately.

For the EM Loupe survey, the team mainly practised the moving-loop walking method. The key training outcomes included two-operator coordination, transmitter–receiver spacing control, line following, steady walking speed, tablet navigation, signal checking, and data saving after each line.

2.3 Phase 2: Costerfield mine survey

2.3.1 Site objective and survey target

The second field survey (Figure 3) was conducted at the Costerfield Mine in central Victoria. Costerfield Mine is an operating underground gold–antimony mine, where mineralisation is primarily associated with narrow, structurally controlled quartz–stibnite–gold veins within the Melbourne Zone; the local stratigraphy consists mainly of Silurian mudstone, siltstone and, locally, sandstone. This mineralisation environment, controlled by a system of faults, shear zones, folds and fractures, makes the mine site suitable for testing near-surface geophysical methods under real-world mining conditions.



Figure 3. Regional location of Costerfield Mine in Victoria, Australia

Permission for this site visit was obtained through liaison with Costerfield Operations. Following approval, the team held pre-survey consultations and site selection discussions with Costerfield Operations' Principal Exploration Geologist. Based on practical access, safety requirements, electromagnetic interference and relevance to the project objectives, Site 3B within the True Blue prospect was identified as the preferred survey area, as shown in Figure 4. This provided a suitable field setting for applying the integrated GPR–EM Loupe workflow under real mine-site conditions.



Figure 4. Study site location within the Costerfield mining area near the True Blue prospect

During the site selection phase, several candidate areas were considered, including areas near the Augusta Mine, Brunswick Mine and the True Blue prospect. These areas differed in accessibility, electromagnetic noise, known mineralisation, drilling activity and geological uncertainty. Site 3B was selected because it offered a useful balance between field practicality and geological value. The area remains relatively underexplored, and the mine operator was interested in its near-surface geophysical response. Existing reverse circulation and diamond drilling data suggest the possible presence of mineralisation beneath the alluvial cover; however, the shallow subsurface structure remains poorly understood. Therefore, Site 3B provided both a suitable geological basis and sufficient uncertainty for testing whether the combined GPR and EM Loupe survey could improve near-surface interpretation.

In accordance with the technical guidelines and field objectives, the team adopted a complementary field deployment strategy combining EM Loupe and GPR. The EM Loupe was used as the primary area-covering method, with survey lines laid out at approximately 10-metre intervals across the 3B grid to map near-surface conductivity variations and identify potential conductive anomalies associated with groundwater, weathering or mineralisation features. Ground-penetrating radar (GPR) was deployed along two boundary lines of the survey area, utilising a 50 MHz low-frequency antenna configuration, to investigate the thickness of alluvial cover, shallow interfaces and possible near-surface structural manifestations by acquiring reflection profiles. Previous Deep Ground Penetrating Radar (DGPR) surveys conducted near Costerfield indicated that radar profiles can identify fault-like, shear-like, contact zone and vein-like features; however, they also revealed that vegetation, tree roots and surface conditions may introduce noise. Consequently, for this survey, priority was given to relatively open pasture areas with few surface obstacles.

2.3.2 Field layout and survey design

Prior to formal data acquisition, the team delineated Site 3B as a rectangular survey area of approximately $180\text{ m} \times 180\text{ m}$, as shown in Figure 5. The survey boundaries, start and end points of survey lines, and intermediate control points were marked using 50 m measuring tapes, steel stakes and marking ropes. Existing LiDAR and imagery data provided topographical and spatial context for the Costerfield area, supporting the understanding of site conditions and the planning of survey lines. However, in this project, these datasets were used mainly as references for field layout rather than as primary interpretative data.



Figure 5. Costerfield practice survey grid layout

The survey line design for this project adopted a strategy of “combining areal coverage with targeted profiles”. EM Loupe served as the primary areal coverage method, with 19 parallel survey lines laid out within the survey area at approximately 10 m intervals to obtain a relatively continuous distribution of near-surface electrical conductivity. During acquisition, the EM Loupe lines were surveyed in alternating directions: odd-numbered lines were walked from east to west, while even-numbered lines were walked from west to east. This helped reduce unnecessary relocation time between adjacent lines and maintain systematic coverage of the grid. GPR data were collected along two selected boundary

profiles, with the first profile acquired from A to B and the second profile acquired from D to C, to provide targeted reflection sections for identifying changes in overburden, shallow interfaces and possible structural discontinuities. The two instruments were not used for simple comparison, but rather as complementary methods: EM Loupe provided areal conductivity information, whilst GPR provided linear structural reflection data. By interpreting both data types within the same survey framework, the project could more reasonably assess whether anomalies were related to variations in cover thickness, water-bearing conditions, shallow structural interfaces or mineralisation-related geological features.

2.3.3 GPR survey method

The GPR survey at Costerfield Mine utilised the Sensors & Software pulseEKKO Pro GPR system. Prior to the formal data acquisition, the team inspected the DVL/control module, transmitter, receiver, fibre-optic cables, 50 MHz antennas, batteries, mounting frame and associated connectors to ensure that battery levels, cable connections, controller displays and system responses were all functioning correctly.



Figure 6. GPR field survey at Site 3B using the pulseEKKO Pro system with a 50 MHz bistatic antenna configuration

Based on the field controller settings, the Site 3B survey utilised a 50 MHz low-frequency bistatic antenna configuration, with key parameters including 2 m transmitter–receiver separation, 0.1 m/ns assumed radar velocity, 400 ns time window, 1.60 ns sampling interval, 250 sampling points, 64 stacks, and 0.5 m antenna step size, and employed the PRO AUTO pulser setting. The data acquisition mode was Reflection Survey. As GPS was not enabled, the spatial positioning of the GPR data relied primarily on pre-marked transects, manual distance measurements and on-site grid controls.

The 50 MHz low-frequency antenna was selected to suit the site conditions at Site 3B. This area features alluvial cover, uncertainties regarding shallow structures, and near-surface anomalies potentially associated with groundwater or mineralisation. Compared to high-frequency antennas, the 50 MHz antenna offers greater penetration but lower resolution, making it more suitable for identifying changes in cover material, shallow interfaces and large-scale structural discontinuities. Based on the field parameters, the theoretical maximum display depth for this GPR survey is approximately 20 m; considering stratigraphic conditions, signal attenuation and data quality, the actual interpretation depth can be conservatively estimated at approximately 18–20 m.

During field acquisition, GPR data were collected along the two boundary lines of the 180 m × 180 m Site 3B survey grid to obtain line-based subsurface reflection profiles, complementing the areal conductivity information acquired by the EM Loupe. To maintain consistent survey geometry, the team

kept the 2 m transmitter–receiver antenna separation as stable as possible and advanced along each survey line using a 0.5 m step interval. This helped reduce geometric errors and positional uncertainty in the subsequent interpretation of the radar profiles.

During data acquisition, the team monitored the radargrams in real time to check for missing traces, excessive noise, unstable antenna contact or abnormal signal responses. If any issues were detected, acquisition was paused to inspect the cables, batteries, trigger settings and antenna positions. Upon completion of the acquisition, the data was promptly backed up and cross-referenced with the corresponding survey line number and field records to support subsequent processing and the integrated interpretation with the EM Loupe conductivity results.

2.3.4 EM Loupe survey method

This project utilises the Loupe Geophysics EM Loupe TEM System to conduct near-surface electrical conductivity surveys. The equipment primarily comprises a transmitter backpack, receiver backpack, control module, 10 m cable, GPS antenna, Ethernet cable, Li-Ion battery, charging cradle and a field tablet. Unlike GPR, which involves the sequential acquisition of reflection profiles along a survey line, the EM Loupe employs walking-mode continuous acquisition. Two operators, each carrying a transmitter and receiver backpack respectively, walk steadily along a pre-marked survey line to continuously acquire near-surface electrical conductivity data.



Figure 7. EM Loupe field survey at Site 3B using transmitter and receiver backpacks

In accordance with the on-site EM Loupe configuration, for this Costerfield survey, the transmitter was set to Loupe mode, with an operating frequency of 375 Hz, a sampling rate of 504,000, and the power feed set to On. The receiver gain was set to 1 in all three directions (X, Y, Z), whilst the remaining parameters were kept at their system default settings. This configuration facilitates the rapid acquisition of information on conductivity variations at Site 3B and ensures consistency in the three-component receiver data.

Regarding line layout, the EM Loupe was utilised as the primary areal coverage tool, collecting data along multiple parallel lines within the 180 m × 180 m Site 3B survey grid, with line spacing of approximately 10 m. This layout provides a relatively continuous near-surface conductivity distribution, which is used to identify conductive anomalies potentially associated with aquifer conditions, clay layers, weathering zones, changes in overburden properties, structural fracture zones or potential mineralisation.

During field acquisition, the team focused on maintaining the relative position of the transmitter and receiver, a consistent walking speed and survey line direction, whilst avoiding fences, vehicles, metallic objects and sources of strong electromagnetic interference as far as possible. During data acquisition, the team used a field tablet to check system status, GPS/tablet records and data continuity; upon completion, the data was saved and backed up by survey line number and cross-referenced with field records for subsequent Loupe DB / EmaxAIR processing, inversion and visualisation analysis.

Overall, EM Loupe primarily provides area-wide conductivity background, whilst GPR provides line-based reflection profiles. The complementary use of both provides a foundation for subsequent interpretation of variations in the overburden, aquifer conditions, shallow geology and anomalies related to potential mineralisation at Site 3B.

2.4 GPR Data processing and interpretation workflow

2.4.1 Project Setup and Basic Processing

A new project was created in EKKO_Project and the raw GPR survey data were imported. Survey lines were renamed as LINE 1 to LINE 11, and a grid from survey Lines was created as shown in Figure 8.

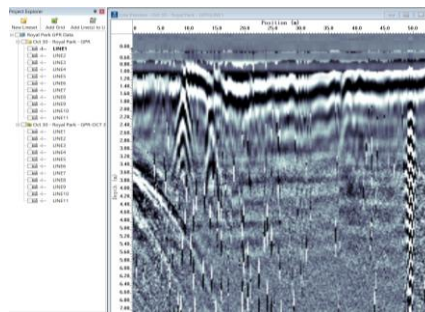


Figure 8. GPR project setup, imported survey lines, and survey layout in EKKO_Project

Raw radargrams were reviewed to assess data quality and identify noise patterns or acquisition inconsistencies. Basic processing steps were done, including time-zero correction, dewow filtering, background removal, gain application and legend settings, as shown in Figure 9.

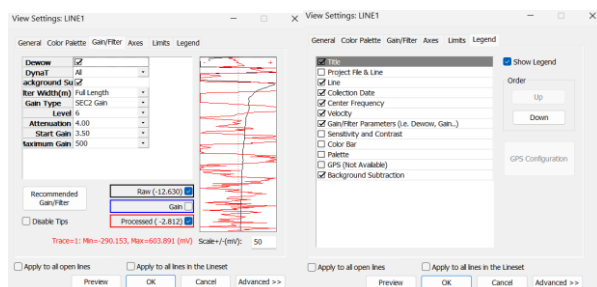


Figure 9. Gain and filter setting, view settings

Initial GPR data review and basic processing steps including time-zero correction, dewow filtering, background subtraction, gain application and legend settings

2.4.2 Wave Velocity

Electromagnetic wave velocity was estimated using hyperbola fitting on selected reflection events, as illustrated in Figure 10. The estimated velocity was subsequently used to convert two-way travel time into approximate subsurface depth.

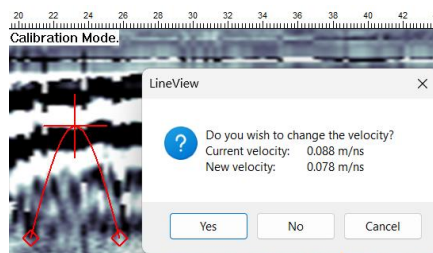


Figure 10. hyperbola fitting

2.4.3 Interpretation of Subsurface Features

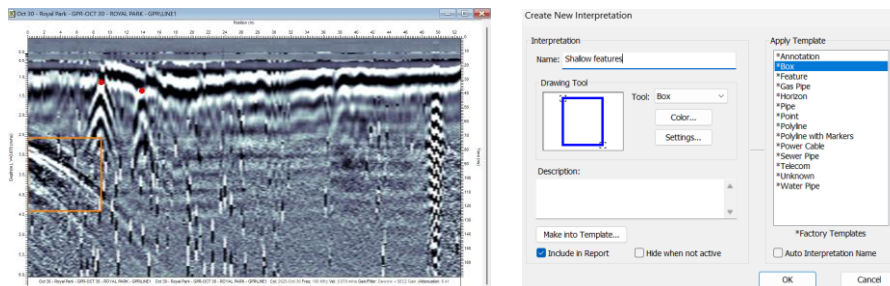


Figure 11 GPR radargram interpretation procedure in EKKO Project, showing the processed radargram and interpretation setup interface

Processed radargrams were interpreted to identify subsurface reflection patterns, layer boundaries, and possible geological anomalies. Reflection continuity, amplitude variation, and geometric characteristics were considered during interpretation. Processed survey lines were also reviewed in map view to examine the spatial relationship of subsurface features across the site.

2.4.4 GPS Coordinates and Depth Slices

GPS coordinates were assigned to the GPR survey grid within EKKO_Project using the Project Position Relationships tool. The origin coordinates and grid rotation relative to north were entered to spatially reference the survey data, as shown in Figure 12.

Depth slices were then generated automatically using the SliceView-Grid function to visualise subsurface reflection patterns at different depths across the survey area. The generated depth slices were subsequently reviewed and processed to improve clarity and identify possible subsurface anomalies and layer variations, as presented in Figure 12.

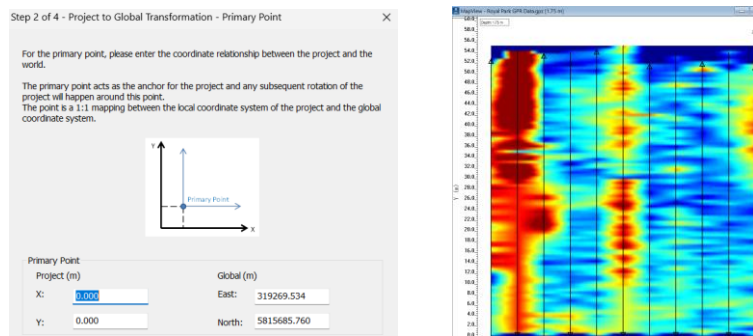


Figure 12. GPS coordinates (left) and Depth Slices (right)

2.5 EM Data Processing

The EM Loupe data were processed using the software workflow consisting of dsQC, EmaxAir, and OriginPro. The workflow involved conversion of the raw EM Loupe data into conductivity-depth imaging (CDI) sections and conductivity depth slices for interpretation of subsurface conductivity variations.

2.5.1 3.5.1 EM Loupe Data Processing Workflow

The EmaxAir GUI software interface is used to convert raw EM Loupe data into a processable format and to perform conductivity inversion; the entire EM Loupe data processing workflow comprises five main stages.

Step 1 .db to .dat

The raw .db files were converted into .dat files using the dsQC software. This conversion reformatted the raw EM Loupe data into a format compatible with EmaxAir processing.

Step 2 .dat to .ldt

The .dat files were converted into .ldt files within EmaxAir. The .ldt format prepared the EM data for conductivity inversion processing.

Step 3 .ldt + .wvf to .par

A parameter (.par) file was generated using the .ldt file, waveform (.wvf) file, and template inversion settings. The .par file defined the conductivity inversion parameters used by EmaxAir.

Step 4 .cdi file

The EmaxAir inversion was executed using the .par file to generate conductivity depth imaging (.cdi) files containing the processed subsurface conductivity data.

Step 5 OriginPro files

The .cdi files were imported into OriginPro for visualisation and interpretation. Conductivity distribution plots, CDI sections, depth slices, and 3D conductivity models were generated for analysis and reporting.

2.5.2 The EM Loupe data processing procedure consisted of five main stages

The generated .cdi files were imported into OriginPro for visualisation and interpretation of subsurface conductivity variations. The imported data columns were renamed according to their survey attributes, including line number, coordinates, depth, conductivity, and elevation information.

Initial conductivity distribution plots were generated to assess the variation of conductivity values across different survey lines. Conductivity-versus-depth plots were then produced using colour-mapped visualisation to identify conductivity trends, noise levels, and the reliable depth of investigation (DOI).

Separate conductivity depth imaging (CDI) sections were generated for individual survey lines using contour-filled plots, where distance was assigned to the X-axis, depth to the Y-axis, and conductivity to the Z-axis. A consistent colour scale was applied to all sections to improve comparison between survey lines, with blue colours representing low conductivity values and red colours representing highly conductive regions.

Conductivity depth slices were also generated at selected depth intervals to visualise the lateral distribution of conductivity anomalies across the survey area. Finally, the depth slices were combined to produce stacked 3D conductivity models for improved interpretation of subsurface conductive features.

2.6 Resources Used for the Investigation

This survey relied primarily on two sets of geophysical equipment: the Sensors & Software pulseEKKO Pro GPR system and the Loupe Geophysics EM Loupe TEM system. The GPR system comprises a 50 MHz antenna, a control/display module, an acquisition unit, fibre-optic cables and batteries, and is primarily used to acquire shallow subsurface reflection profiles. The EM Loupe system comprises a transmitter backpack, a receiver backpack, a tablet control interface, connecting cables and a battery, and is primarily used to acquire near-surface electrical conductivity mapping data and conductivity-depth information.

Field deployment and recording were primarily carried out using manual measurements and field documentation tools. The team utilised tape measures, steel stakes, wooden stakes, marking ropes and compasses to define survey boundaries, line positions and control points. A field logbook was used to record line numbers, acquisition sequences, field observations, equipment status and issues encountered during data acquisition. Photographs were taken to document site layout, equipment operation and survey conditions. Upon completion of data acquisition, the team backed up the raw data using a laptop to mitigate the risk of data loss.

Costerfield Mine provided a real-world mining environment for the final survey. Survey Area 3B enabled the team to apply the fieldwork procedures developed during the training phase at Royal Park to actual mining conditions, thereby testing the operational feasibility of GPR and EM Loupe in a real-world mining context. In addition to field equipment, this project required corresponding software support, including EKKO Project for GPR data viewing and processing, and Loupe DB / EmaxAIR for

EM Loupe data management and processing. Furthermore, the successful execution of the project relied on technical guidance from the supervisor, division of labour and collaboration among team members, site access and coordination support provided by mine personnel, as well as the provision of necessary PPE, safety training and site access permits to ensure that fieldwork was conducted safely and in compliance with regulations.

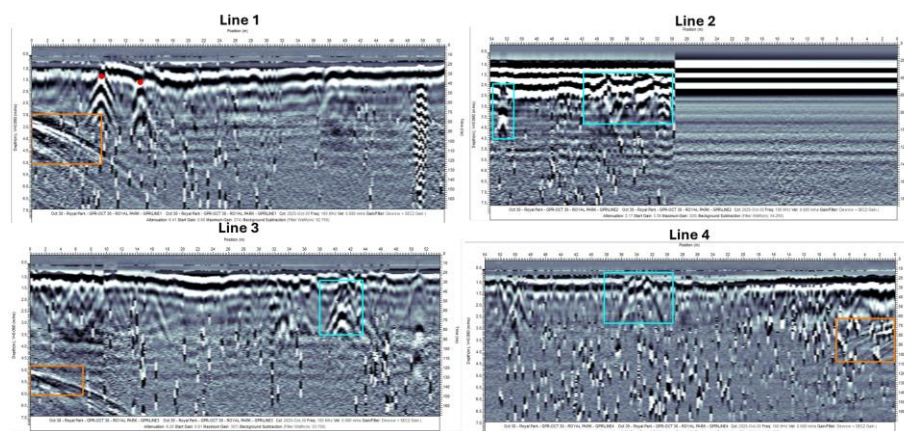
3 Result and Analysis

3.1 Geophysical survey findings of Royal Park

3.1.1 GPR data analysis for Royal Park

3.1.1.1 GPR interpretation

The Royal Park GPR survey lines (Figure 13) generally show limited subsurface penetration, with most identifiable features occurring within the upper 1–3 m. Across Lines 1–11, several shallow anomalous reflections were identified and highlighted using light blue boxes. These features appear as irregular and discontinuous reflections, which are interpreted as near-surface noise or interference rather than genuine subsurface targets. Incomplete hyperbolas are marked using orange boxes, while complete or near-complete hyperbolas are indicated using red dots. A GPR wave velocity of 0.08 m/ns was adopted for the site, consistent with typical moist soil, saturated silt, and clay-rich ground conditions. Using this velocity and the selected 100 MHz antenna frequency, the approximate maximum depth of investigation was around 7 m.



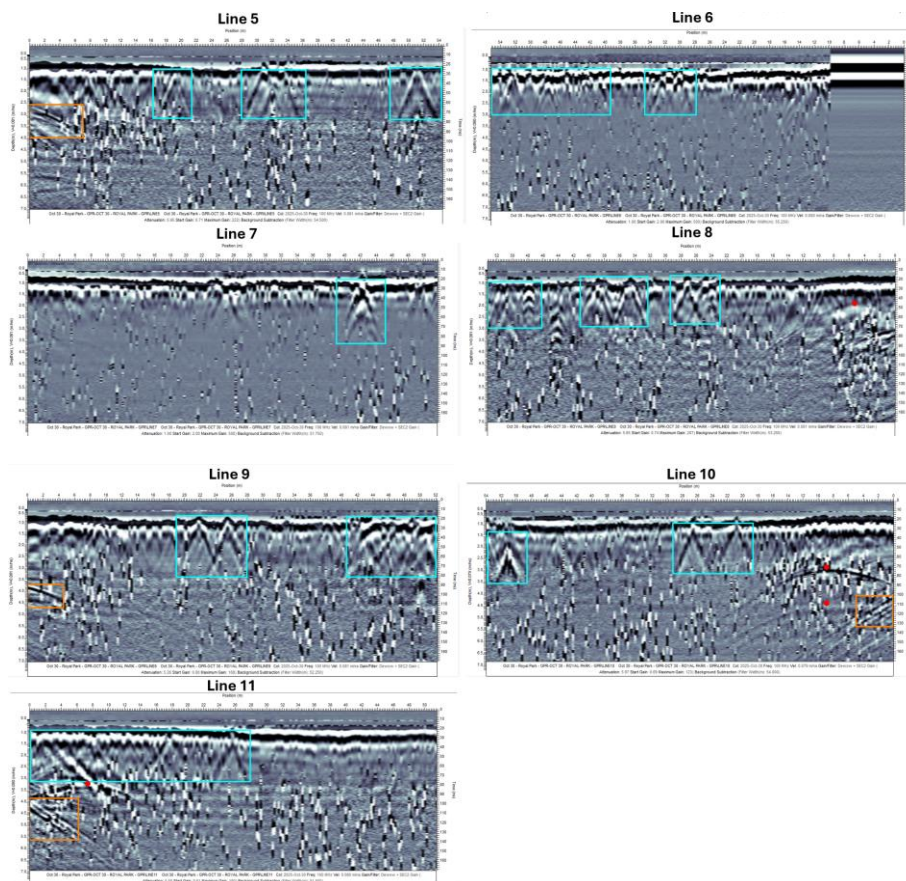


Figure 13. GPR line profile

In Line 1 (Figure 13), two small hyperbolas are identified at approximately 1.3 m and 1.5 m depth, with an estimated velocity of 0.067 m/ns. This velocity is most consistent with saturated silt (0.055–0.064 m/ns) and gravel (0.072–0.076 m/ns), suggesting moist or saturated near-surface materials.

In Line 8, a distinct hyperbola is identified at approximately 1.7 m depth, with an estimated velocity of 0.281 m/ns, as shown in Figure 13. The calculated velocity is close to the electromagnetic wave velocity in air (~0.300 m/ns), indicating that the feature is likely caused by air wave noise rather than a buried subsurface target. Similarly, in Lines 10 and 11, as shown in Figure 13, large hyperbolas are observed at approximately 3 m and 3.2 m depth, respectively. The estimated velocities are 0.263 m/ns and 0.250 m/ns, which are both close to the air wave velocity, suggesting these features are likely caused by air wave noise rather than buried subsurface targets.

Therefore, this feature is similarly interpreted as air wave interference or near-surface noise rather than a genuine subsurface object. The complete hyperbolas are mainly observed in Lines 1, 8, 10, and 11. Hyperbola velocity analysis in EKKO Project produced estimated velocities ranging from

approximately 0.250 to 0.281 m/ns, which are close to the electromagnetic wave velocity in air (~0.300 m/ns). Therefore, these reflections are interpreted as air wave noise rather than buried objects or geological features.

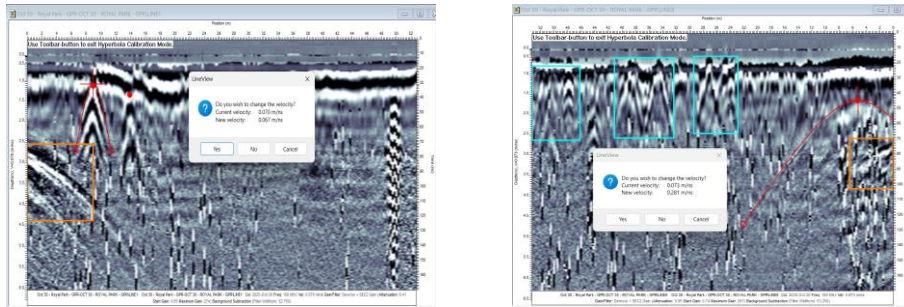


Figure 14 . GPR wave velocity calibration for Royal Park Line 1 (left) and Line 8 (right).

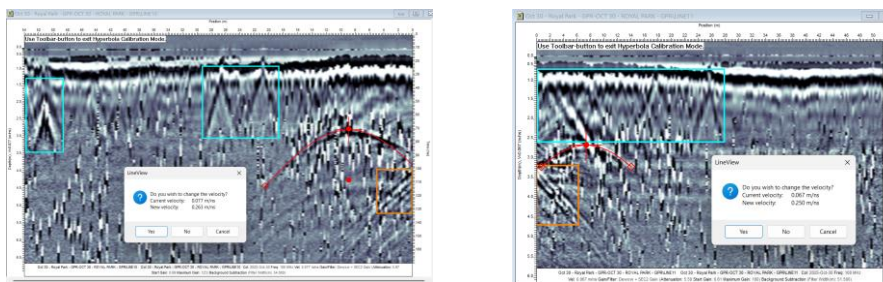


Figure 15. GPR wave velocity calibration for Royal Park Line 10 (left) and Line 11 (right).

No clear soil layer boundaries were identified within the GPR profiles. Considering the selected 100 MHz antenna frequency, the approximate depth of investigation was only around 6.5 m. Under these conditions, weak dielectric contrasts, signal attenuation, and conductive ground conditions may prevent clear identification of subsurface stratigraphy. Therefore, the absence of distinct soil boundary reflections is considered reasonable for this survey.

3.1.1.2 GPR depth slices for Royal Park

The generated depth slices indicate strong subsurface reflections within the shallow depth range of approximately 1 m to 2 m, particularly around survey Lines 2 and 6, as shown in Figure 16. The strongest reflections occur at approximately 1.5 m depth, where high-amplitude zones are clearly visible across parts of the survey grid. These strong reflections may indicate significant material contrasts, moisture variations, or possible subsurface anomalies within the shallow subsurface.

At depths of approximately 3 m to 4 m, the reflection amplitudes become noticeably weaker, suggesting increased signal attenuation or a more homogeneous subsurface layer. However, reflections reappear between approximately 5 m and 7 m depth, although these reflections are relatively weak, scattered, and discontinuous across different survey lines. This pattern may suggest deeper isolated reflection zones or weaker subsurface interfaces rather than a continuous geological boundary.

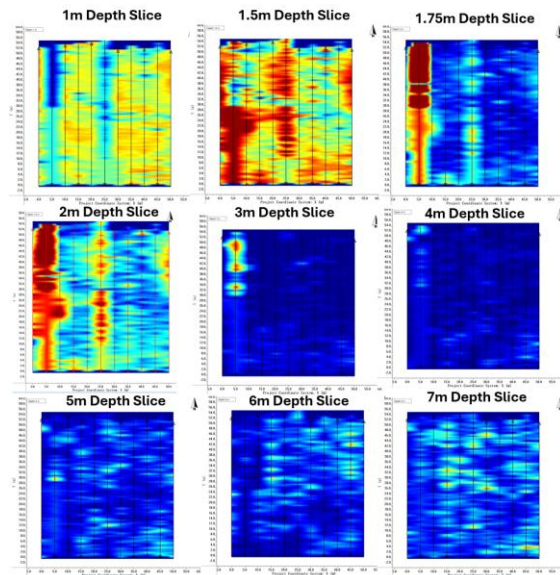


Figure 16 Grid depth slices showing obvious subsurface variations

3.1.2 EM data analysis for Royal Park

In the contour point distribution shown in Figure 17, there are no data points within the 0–8 m depth range, while consistent conductivity data points are observed mainly below approximately 10 m depth. The data density decreases significantly below approximately 30 m depth, with increasingly sparse and scattered conductivity points observed at greater depths. Therefore, a depth of investigation (DOI) of 30 m was adopted for the EM Loupe dataset, and the interpretation focused mainly on conductivity responses above 30 m depth.

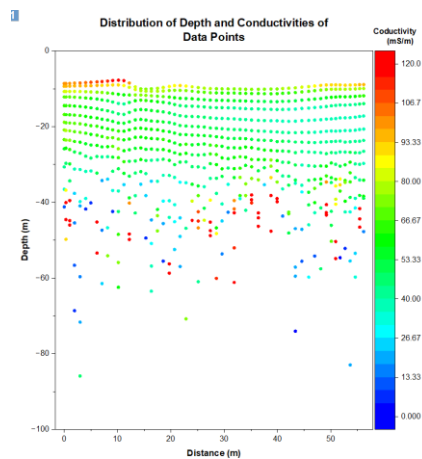
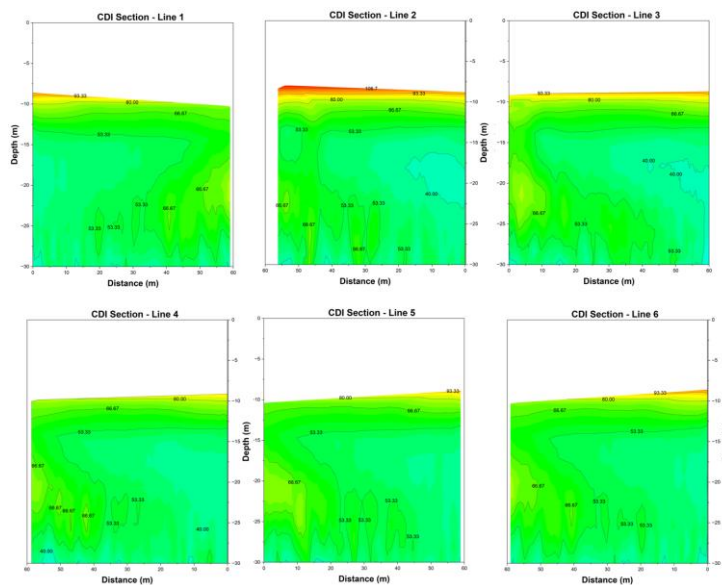


Figure 17. Distribution of depth and conductivities of data points for line 1 in Royal Park

Figure 18 shows the conductivity-depth imaging (CDI) sections for survey Lines 1–11. Overall, the CDI sections show similar conductivity distributions, with most conductivity values ranging between 40–80 mS/m.

The shallow layer (8m-12m) with conductivity values of approximately 66.67–80 mS/m is most consistent with wet clay, saturated silt, or moist fine-grained soil. Wet clay typically has conductivity values ranging from 20–1000 mS/m, while saturated silt can have conductivity values up to 100 mS/m. This interpretation is consistent with the wetland-influenced conditions and weathered basaltic clay described in the site geology.

The deeper layer (12-30m) with conductivity values of approximately 40–53.33 mS/m is also consistent with fine-grained soil and saturated silt. Fine-grained soil typically has conductivity values around 40 mS/m, while saturated silt commonly ranges between 1–100 mS/m. The slightly lower conductivity at greater depths may indicate reduced moisture content or transition into sandier materials.



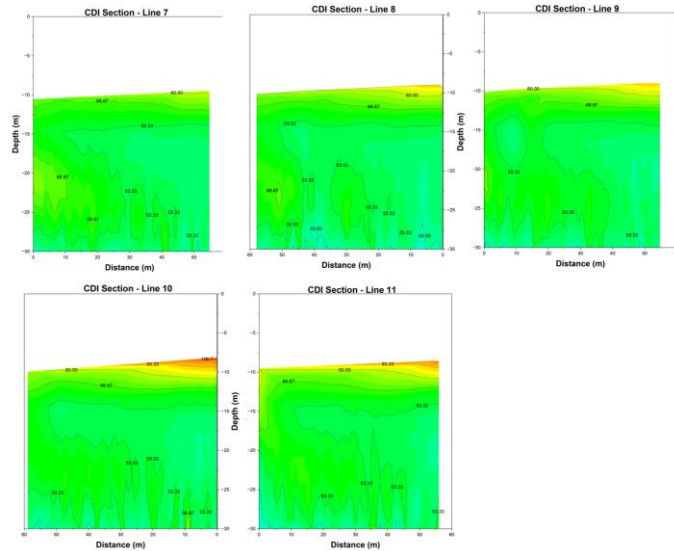


Figure 18 Conductivity-depth sections of Royal Park

3.1.2.1 EM Loupe depth slices for Royal Park

Figure 19 shows the conductivity depth slices between 10–30 m depth. Overall, most areas remain within a conductivity range of approximately 53.33–66.67 mS/m across all depth slices. A lower conductivity zone of approximately 40 mS/m is observed in the south-west corner at the 15–20 m depth slice. This lower conductivity region remains visible in the 20–25 m depth slice but becomes significantly smaller in the 25–30 m depth slice. Based on the conductivity values, this area is most consistent with less conductive fine-grained soil, sandy material, or sandstone-rich material, which typically exhibit lower conductivity compared to moist clay-rich soils.

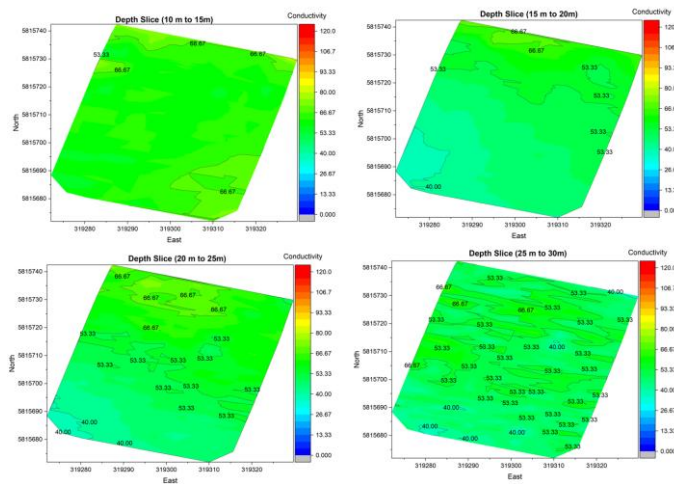


Figure 19 Apparent conductivity map of Royal Park

3.2 Geophysical Survey Findings of Costerfield Mine

3.2.1 GPR data analysis for Costerfield Mine

Based on the geological map (Figure 20) provided by the principal geologist from the mining site, the investigation area is categorised as Costerfield Siltstone (Slc), consisting mainly of thin-bedded siltstone with minor sandstone interbeds. This indicates that the subsurface is predominantly composed of fine-grained sedimentary rock with local sandstone layers. Based on these geological conditions, a GPR wave velocity of 0.078 m/ns was adopted for the Costerfield survey, representing an average velocity for common subsurface materials in the area such as silt and sand.

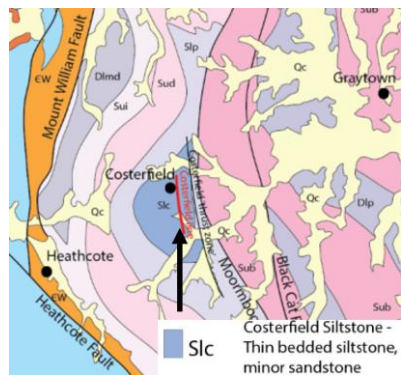


Figure 20 Geological map of the Costerfield survey area.

Line profiles were obtained and interpreted based on characteristic features visible in the radargrams, as shown in Figure 21 for Line AB and Line CD.

In Line AB (Figure 21), several large hyperbolic reflections (orange boxes) are visible between approximately 1.5 m and 12.5 m depth. As shown in Figure 21, the fitted velocities of these hyperbolas range between approximately 0.25–0.30 m/ns, which are significantly higher than typical soil velocities and are therefore interpreted as air waves or antenna ringing effects rather than true subsurface targets. A relatively continuous straight-line reflector is also observed at approximately 6.5 m depth, which likely represents a boundary between subsurface materials with contrasting dielectric properties, interpreted as a soil or lithological boundary.

In Line CD (Figure 21), the highlighted features appear as steeply inclined and vertically disrupted reflections that cut across the surrounding relatively horizontal and continuous layers. These are interpreted as three shallow fault-like features within the yellow boxes between approximately 2.2 m and 11.5 m depth. However, GPR data alone cannot conclusively confirm geological faults without supporting geological or borehole information. In addition, a curved reflector highlighted in light blue is observed at approximately 13.7 m depth, which is interpreted as a subsurface material boundary or gradual change in geological layering.

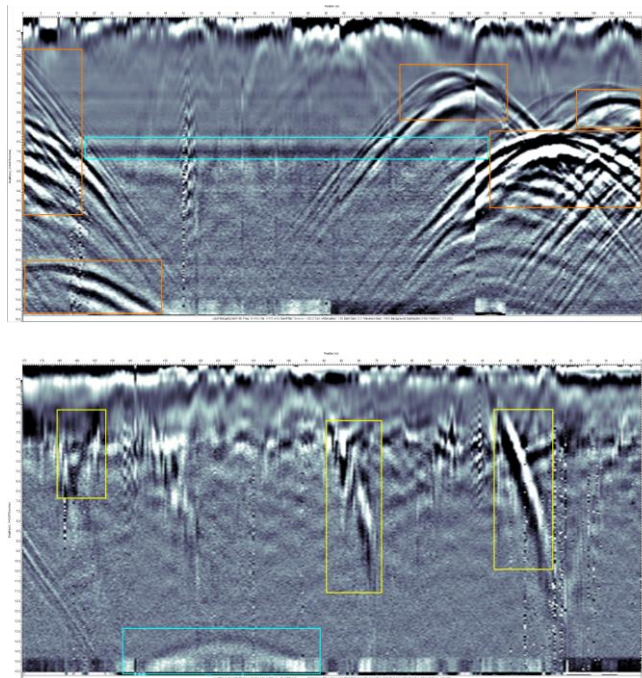


Figure 21. GPR interpretation results for Line AB (top) and Line CD (bottom)

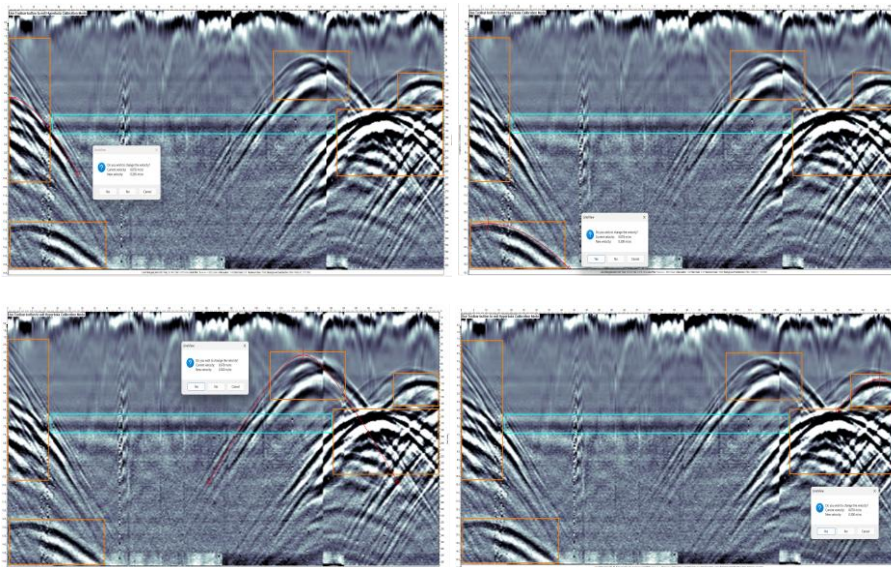


Figure22. GPR Velocity of Hyperbolas in Line AB

3.2.2 EM Loupe data analysis for Costerfield mine

Figure 23 shows the distribution of conductivity data points with depth for the Costerfield EM Loupe survey. The data density remains relatively consistent down to approximately 50 m depth. Below this depth, the conductivity data become significantly more scattered and sparser, indicating reduced data reliability. Therefore, a depth of investigation (DOI) of approximately 50 m was adopted for the EM Loupe survey setup at Costerfield, and interpretations mainly focused on conductivity responses above 50 m depth.

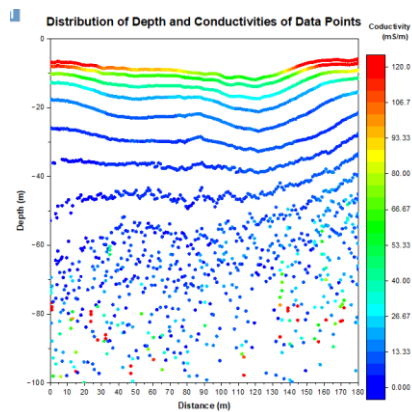


Figure 23 Distribution of Depth and Conductivities of Data Points for line 1

Figure 24 shows the conductivity-depth imaging (CDI) sections for survey Lines 1–18 at Costerfield. Overall, the survey lines show a similar conductivity pattern, with conductivity decreasing progressively with depth. Conductivity values are generally above 100 mS/m at approximately 8 m depth, decreasing to around 80 mS/m at 10 m depth, approximately 40 mS/m at 14 m depth, and below 25 mS/m beneath approximately 18 m depth.

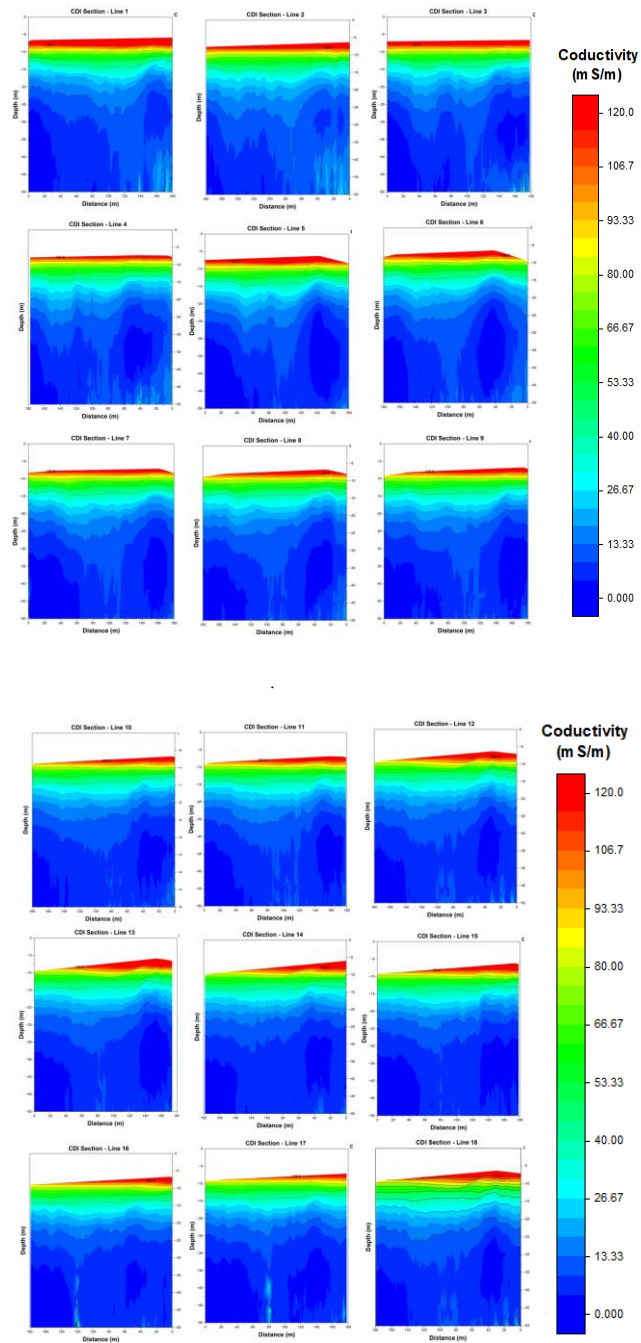


Figure 24 Conductivity-depth sections

3.2.2.1 EM Loupe depth slices for Royal Park

Figure 25 shows the conductivity depth slices for the Costerfield survey, representing the lateral conductivity distribution at different depths. At 7–10 m depth, relatively high conductivity values (>100 mS/m) are observed within the south-western one-third and north-eastern corner of the survey area. These high conductivity zones are most consistent with moist clay-rich or saturated fine-grained material, which commonly exhibit elevated conductivity values due to higher moisture content.

Below 10 m depth, conductivity variation across the survey area becomes relatively uniform, with less than approximately 15 mS/m variation between different locations. At 10–15 m depth, most areas exhibit conductivity values between approximately 40–55 mS/m. Between 15–20 m depth, conductivity decreases to approximately 25 mS/m in most areas, with some isolated zones below 15 mS/m. From 20–50 m depth, conductivity values are generally around 15 mS/m and remain mostly below 30 mS/m throughout the survey area. These lower conductivity responses are more consistent with less conductive siltstone and minor sandstone material within the Costerfield Siltstone (Slc) unit.

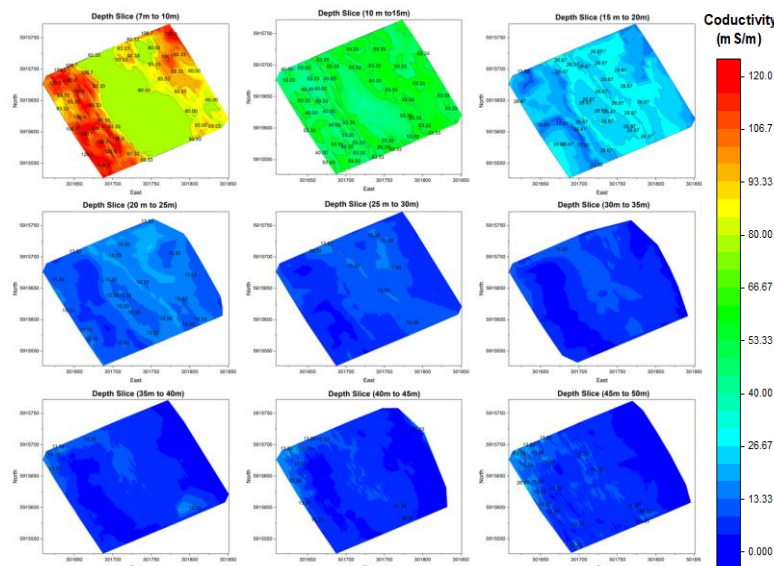


Figure 25 EM Loupe depth slices for Royal Park

3.3 Novel Contributions

This project demonstrates the integrated application of Ground Penetrating Radar (GPR) and EM Loupe transient electromagnetic surveying for near-surface investigation within an active gold–antimony mining environment at Costerfield Mine. The investigation combined shallow GPR reflection profiling with EM Loupe conductivity-depth imaging and conductivity depth slices to improve interpretation of shallow subsurface conditions beneath alluvial cover. The integrated interpretation identified possible shallow interfaces, conductive near-surface layers, fault-like reflection disturbances and deeper low-conductivity zones consistent with the regional Costerfield Siltstone geological setting.

The project also developed and applied a complete field-to-processing workflow under real mine-site conditions. This included field survey planning, GPR and EM Loupe acquisition, field quality control, GPR processing in EKKO Project, EM Loupe inversion using EmaxAIR, and conductivity visualisation and interpretation in OriginPro. The combined use of line-based GPR interpretation and area-wide EM

conductivity mapping provided complementary subsurface information and demonstrated the practical application of integrated near-surface geophysical methods for preliminary mining investigations.

4 Discussion

4.1 Integrated GPR and EM Loupe Interpretation and Conclusion

The findings and interpretation from the integrated GPR and EM Loupe survey are summarised in Table 1. Soil and rock layers were identified based on changes in conductivity values together with soil boundaries detected from the GPR interpretation, as shown in Figure 26. The interpreted subsurface layering was further compared with the local geological setting shown in Figure 20, which identifies the area as Costerfield Siltstone (Slc), consisting mainly of thin-bedded siltstone with minor sandstone interbeds.

Possible subsurface materials corresponding to the observed conductivity ranges were identified using published conductivity reference values, including silt, saturated silt, wet clay, shale, sand, sandstone, and gravel-rich material. The interpretations were made based on the combined analysis of conductivity magnitude, conductivity variation with depth, GPR reflection characteristics and local geology.

Table 1. Integrated GPR and EM Loupe interpretation of subsurface layers and possible materials at the Costerfield survey area.

Layer	Depth range	EM conductivity range	GPR observation	Possible materials based on conductivity	Interpretation
Layer 1	0–6.5 m	No reliable EM data	Shallow fault-like features detected by GPR	No EM data	Shallow disturbed or fractured/weathered zone within Costerfield Siltstone
Layer 2	6.5–10 m	80–130 mS/m	Soil boundary detected near 6.5 m	Wet clay: 20–1000 mS/m; saturated silt: ≤ 100 mS/m; wet shale: 10–100 mS/m; dry clay: 2–100 mS/m	Highly moist clayey/weathered siltstone or saturated fine-grained zone
Layer 3	10–14.5 m	40–80 mS/m	Soil boundary detected near 14.5 m	Wet clay: 20–1000 mS/m; saturated silt: ≤ 100 mS/m; unsaturated silt: 1–100 mS/m; wet shale: 10–100 mS/m; fine-grained soil: ~ 40 mS/m	Moist siltstone with clay-rich or saturated fine-grained intervals
Layer 4	14.5–17 m	25–40 mS/m	No GPR data	Unsaturated silt: 1–100 mS/m; saturated silt: ≤ 100 mS/m; wet shale: 10–100 mS/m; dry clay: 2–100 mS/m; fine-grained soil: ~ 40 mS/m	Transition from moist/weathered siltstone to lower-conductivity siltstone
Layer 5	17–50 m	< 25 mS/m	No GPR data	Unsaturated silt: 1–100 mS/m; dry clay: 2–100 mS/m; wet shale: 10–100 mS/m; wet sand: 0.1–10 mS/m; saturated sand and gravel: 0.7–9 mS/m	Less conductive Costerfield Siltstone with minor sandstone-rich or more competent rock zones

Figure 26 shows a clear decrease in conductivity with depth across the survey area. The shallow layers between approximately 6.5–14.5 m depth exhibit relatively high conductivity values, indicating moist and clay-rich fine-grained material within the upper weathered siltstone sequence. These conductive layers correspond closely with the conductivity anomalies identified in the CDI sections and depth slices.

Below approximately 17 m depth, conductivity values decrease significantly and remain relatively uniform throughout the deeper profile. This lower conductivity response is consistent with less weathered Costerfield Siltstone containing minor sandstone interbeds. The similar conductivity patterns

observed across multiple CDI sections indicate that the subsurface layering is relatively continuous across the survey area.

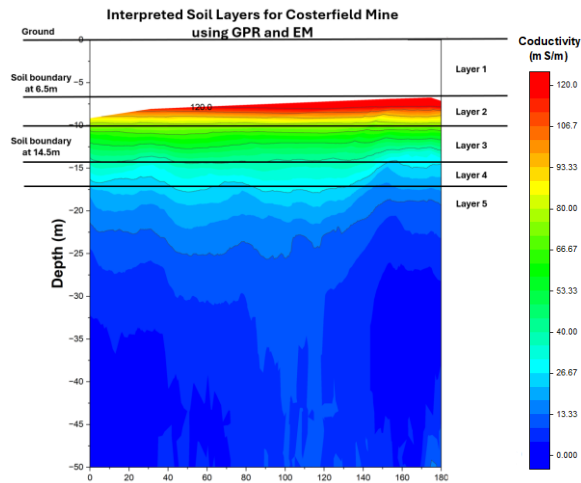


Figure 26. Interpreted subsurface conductivity layers for CDI Section – Line 9 using integrated GPR and EM Loupe interpretation.

4.2 Interpretation of Results within the Context of Existing Literature

The GPR results obtained at the Costerfield survey site are generally consistent with previous studies reviewed in the literature. Existing research indicates that GPR is effective for identifying shallow interfaces, fractures, structural discontinuities and changes in overburden within mining and quarry environments (Francke, 2012; Caselle et al., 2020; Rey et al., 2015). In this project, the relatively continuous reflector identified at approximately 6.5 m depth in Line AB is interpreted as a possible subsurface boundary between shallow conductive overburden and underlying weathered siltstone. The steeply inclined and vertically disrupted reflections observed in Line CD are also consistent with the type of fault-like or fractured features described in previous GPR studies of hard-rock environments (Rey et al., 2015). In addition, the large hyperbolic reflections interpreted as air-wave interference demonstrate limitations discussed in the literature, where surface conditions, noise and antenna effects may influence GPR interpretation (Davis & Annan, 1989; Francke, 2012).

The EM Loupe results also align with findings presented in previous TEM and near-surface conductivity studies. Existing literature shows that transient electromagnetic methods are effective for identifying conductivity variations associated with groundwater, clay-rich material, weathering and geological heterogeneity (Van Dam et al., 2020; Soares & Wijns, 2023; Sánchez Mora et al., 2024). At the Costerfield survey site, the shallow conductive zone between approximately 7–10 m depth, with conductivity values commonly above 100 mS/m, is interpreted as moist clay-rich or saturated fine-grained material, while the lower conductivity values at greater depths are consistent with less weathered siltstone and sandstone-rich material. The integrated interpretation of GPR reflectors and EM conductivity variations also supports conclusions from previous studies that combining multiple geophysical methods can improve interpretation reliability compared with relying on a single dataset alone (Sánchez Mora et al., 2024).

Overall, the results of this project are consistent with the literature review, which identified GPR as suitable for shallow structural characterisation and EM Loupe as effective for broader conductivity mapping and subsurface material variation assessment.

4.3 Limitations

4.3.1 Fieldwork and survey coverage limitations

The Costerfield survey was affected by several fieldwork constraints. Rain reduced the available survey time and indirectly limited the GPR survey to only two boundary lines, A–B and D–C. As a result, the GPR dataset provides useful line-based information but cannot fully represent the whole 180 m × 180 m investigation area. The site surface also included grass, tree roots, muddy areas, uneven ground and local vegetation, which may have affected walking stability, antenna movement and line-following accuracy. For the EM Loupe survey, 19 lines were originally planned, but one line was lost due to a data recording issue. The remaining 18 lines still support broad conductivity interpretation, but the missing line reduces local continuity and may affect interpolation between adjacent lines. Therefore, the survey should be considered a preliminary site-scale investigation rather than a complete spatial model of the subsurface.

4.3.2 Instrument and data-quality limitations

The GPR data were generally clear, and continuous reflectors could be observed. However, the 50 MHz antenna was selected to improve penetration depth, which inherently reduces near-surface resolution compared with higher-frequency antennas (Davis & Annan, 1989). As a result, thin layers, small fractures and weak reflectors may not be clearly resolved. Signal attenuation was also observed at depth, and deeper reflections are generally more difficult to interpret because radar energy decreases with increasing travel distance and material losses (Sensors & Software Inc., 2022). In addition, no CMP survey, borehole tie or site-specific velocity calibration was carried out. The GPR depth conversion therefore relied on an assumed radar velocity, introducing uncertainty into the estimated reflector depths (Davis & Annan, 1989).

The EM Loupe data were generally stable, with no major noise or abnormal data jumps. However, the dataset did not provide useful information in the shallow 0–7 m range, which is consistent with the depth sensitivity characteristics of the system (Loupe Geophysics Pty Ltd., 2024). The main interpretation depth at Costerfield was limited to approximately 50 m, while data below this depth became increasingly sparse and uncertain due to reduced sensitivity at greater depths (Loupe Geophysics Pty Ltd., 2024). Conductivity anomalies also have non-unique causes, which is a common limitation of electromagnetic methods (Reynolds, 2011). High conductivity may be associated with groundwater, clay-rich materials, weathering products, salinity or mineralisation-related materials (Reynolds, 2011). Therefore, EM Loupe anomalies should not be interpreted directly as mineralisation without supporting geological or drilling evidence (Loupe Geophysics Pty Ltd., 2024; Reynolds, 2011).

4.3.3 Validation limitations

A major limitation of this project is the lack of direct subsurface validation. No borehole logs, test pit data or direct sampling results were available along the survey lines. The mine provided basic topographic and site information, and field geological observations were made during the survey. However, these sources cannot directly confirm the cause of each GPR reflector or EM conductivity anomaly. Therefore, the interpretation cannot confidently assign a specific reflector to a geological layer,

fault, groundwater boundary or mineralised zone. All interpreted features should be described using cautious terms such as “possible”, “likely” or “may indicate”. This limitation is important because geophysical data are indirect and require ground truthing to support reliable geological interpretation (Reynolds, 2011).

5 Conclusion

This project investigated the use of Ground Penetrating Radar (GPR) and EM Loupe transient electromagnetic surveying for near-surface geophysical investigation at the Costerfield survey site. The study successfully completed field acquisition, data processing and integrated interpretation using GPR reflection profiles, EM conductivity-depth imaging and conductivity depth slices. The results identified shallow reflection boundaries, possible fault-like disturbed zones, conductive near-surface layers and deeper low-conductivity zones across the survey area. The integrated interpretation indicates that the shallow subsurface is likely dominated by weathered and moisture-rich fine-grained material above less weathered Costerfield Siltstone containing minor sandstone interbeds.

The project objectives were achieved through the combined interpretation of GPR and EM Loupe datasets. GPR provided relatively high-resolution information on shallow reflectors and structural disturbances within approximately the upper 10 m, while EM Loupe provided broader conductivity information to depths of approximately 50 m. The correlation between GPR reflection boundaries and EM conductivity variation improved confidence in the interpretation of shallow subsurface layering and demonstrated the value of integrating both methods for preliminary mining investigations. However, the geophysical data alone is not sufficient to directly confirm subsurface conditions or geological materials. The interpreted layers, conductivity zones and fault-like features remain preliminary because no direct ground-truth validation was available.

Future work should focus on improving interpretation confidence through additional subsurface validation and expanded survey coverage. Borehole data, drillhole logging, trenching and direct geological sampling along the survey lines would allow direct comparison between the geophysical responses and actual subsurface conditions. Additional GPR survey lines and denser EM Loupe coverage could improve spatial continuity and reduce uncertainty between survey lines. The use of multiple GPR antenna frequencies may also improve both shallow resolution and deeper penetration, allowing better identification of thin layers, fractures and structural boundaries.

Further work could also include repeated surveys under different moisture conditions to evaluate the influence of groundwater and seasonal variation on conductivity response. In addition, combining geophysical surveys with existing mine geological information and future drilling programs would improve interpretation reliability and support more detailed geological assessment. The combined use of non-intrusive geophysical methods and intrusive investigation techniques has strong potential to provide more reliable and useful subsurface information for future mining investigations at the Costerfield survey site.

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8 Declaration

Acknowledgement of tools used: We acknowledge the use of ChatGPT, an AI language model developed by OpenAI, during the preparation of this report.

How the tool was used: ChatGPT was used to support language editing, grammar checking, refinement of academic expression, report structure checking, and the development of draft wording for selected sections. It was also used to help clarify how to respond to supervisor feedback and to improve the readability and logical flow of the report. The AI tool was not used to collect field data, operate the GPR or EM Loupe equipment, conduct the original data acquisition, or independently determine the final geological interpretation. All technical content, field observations, data processing decisions, results interpretation and final conclusions were reviewed, checked and approved by the project team.

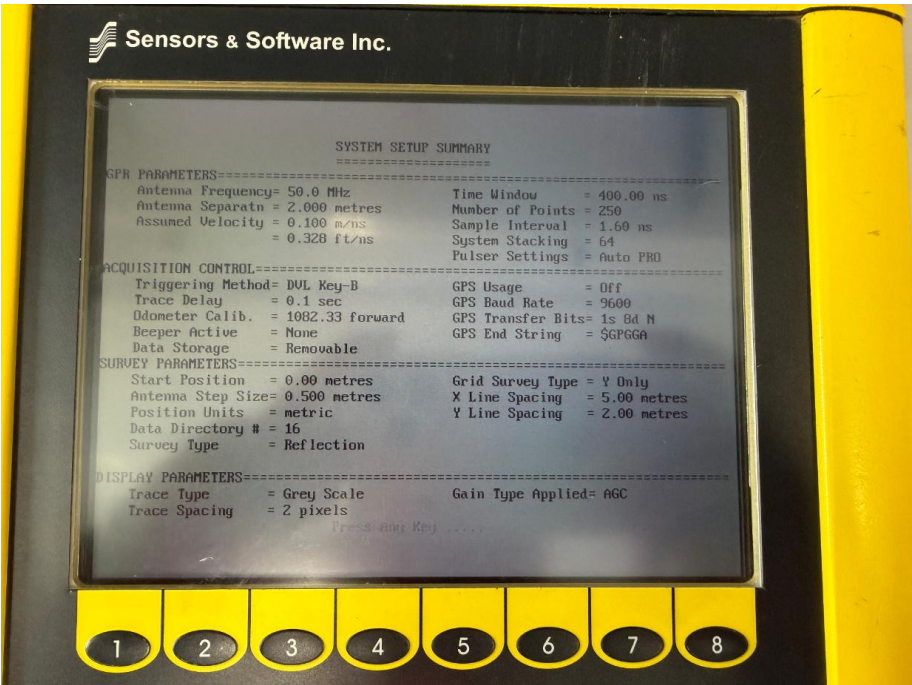
9 Appendices

Appendix A EmaxAir Gui



Appendix C GPR

Appendix C.1 GPR system settings



Appendix D EM Setups

